

UNIVERSITÄT
LEIPZIG

The Effect of Visualization on Mental Models of Polarization

Jonas Bley, Judith Schmid, Fabian Hennig, Malte Ubben, Philipp Bitzenbauer

Professur für Physikdidaktik

Preregistration: osf.io/u6xk9

Overview

1. Mental models and model understanding
2. The polarization applet
3. Hypotheses and study design
4. Analysis and current status

Mental models and model understanding

UNIVERSITÄT
LEIPZIG

Gestalt and Functionality

Gestalt

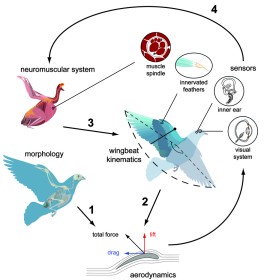
Forms, spatial arrangements, and structures in a learner's mental representation.

Functionality

Relationships and operations used to explain behavior and make predictions.

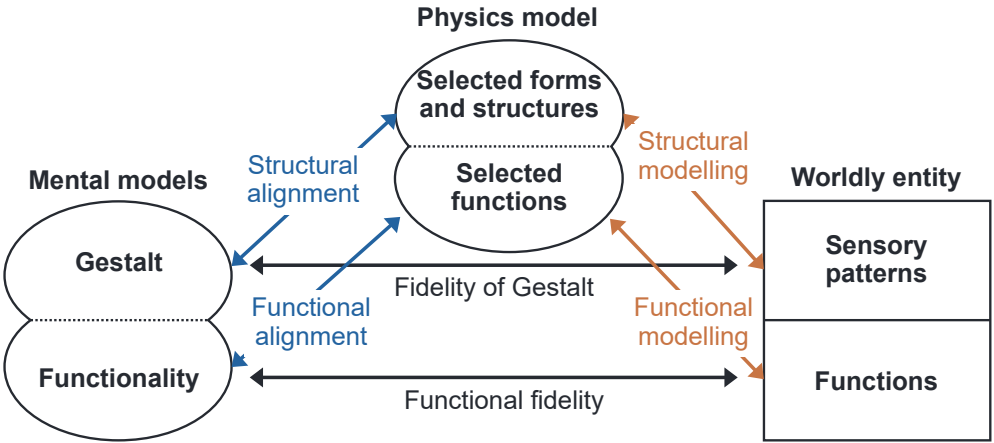


(WikimediaCommons, 2024)



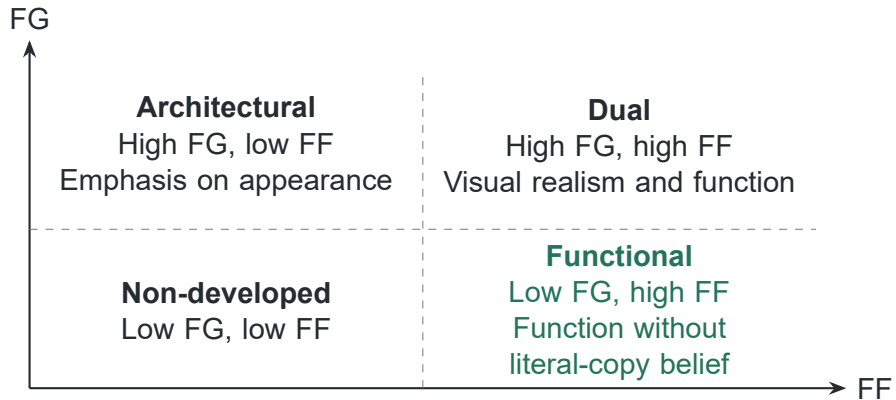
(Altshuler et al., 2015)

Mental models, scientific models, and the world



Collaborative work by Judith Schmid, Malte Ubben, Fabian Hennig, Jonas Bley, and Philipp Bitzenbauer (unpublished, 2026). Black: fidelity; blue: expert alignment; orange: scientific modelling.

Four profiles of model understanding



(Ubben and Bitzenbauer, 2022).

Fidelity and alignment answer different questions

Fidelity: the learner's stance

- What does the learner believe the model is faithful to?
- FG: appearance; FF: function.

Alignment: expert correspondence

- **Structural alignment (StA):** correspondence with expert-selected forms and structures.
- **Functional alignment (FA):** correspondence with expert-selected functions.

High structural alignment can coexist with high FG.

Visualization and the multimedia effect

Multimedia principle

Combining words with relevant pictures can benefit learners more than words alone.

(Mayer, 2005)

DeFT: Design, Functions, Tasks

- **Design:** Which representations are provided, how information is distributed, and how the translations are highlighted.
- **Functions:** Representations can **complement** each other, **constrain** interpretations, and support the construction of **deeper understanding**.
- **Tasks:** Learners must understand each representation and relate its information to the others.

(Ainsworth, 2006)

Two pathways for an additional visualization

Representational support

- Make state relationships visible.
- Connect notation, geometry, and optical transformations.
- Support expert-aligned explanations and predictions.

Spontaneous reflection

- Encounter different representations of the same state.
- Reconsider literal interpretations.
- Attend to shared functions and limits.

Due to the multimedia principle, expert alignment may improve even when fidelities remain stable.

The polarization applet

Jones vectors and the Poincaré sphere

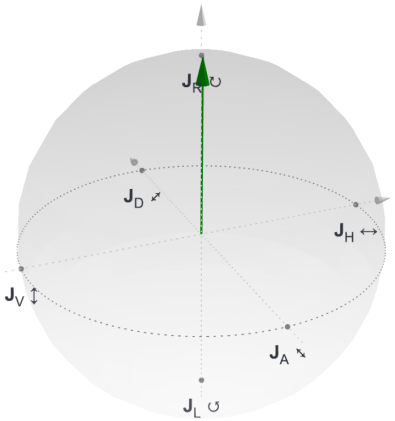
For a normalized, fully polarized state,

$$\mathbf{J} = \begin{pmatrix} a \\ b\, e^{i\varphi} \end{pmatrix}, \qquad a, b \geq 0, \quad a^2 + b^2 = 1.$$

The applet uses

$$\mathbf{S} = \begin{pmatrix} a^2 - b^2 \\ 2ab\cos\varphi \\ 2ab\sin\varphi \end{pmatrix}, \qquad \|\mathbf{S}\| = 1,$$

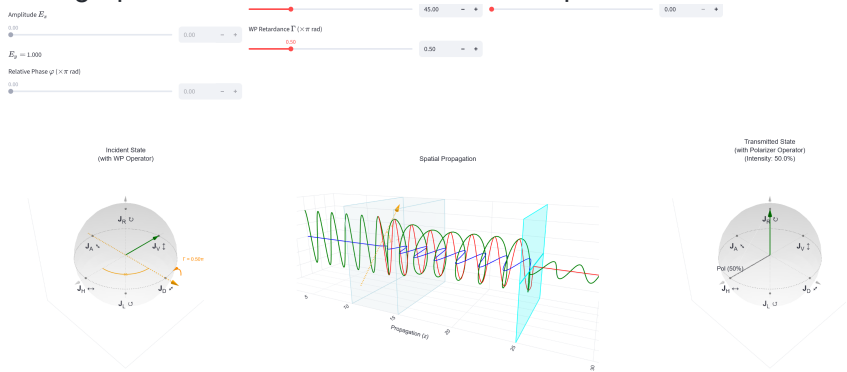
akin to Qubit on the Bloch sphere
$|\psi\rangle = \cos\frac{\vartheta}{2}|0\rangle + e^{i\varphi}\sin\frac{\vartheta}{2}|1\rangle$



The applet

Control: Jones notation and spatial wave graph.

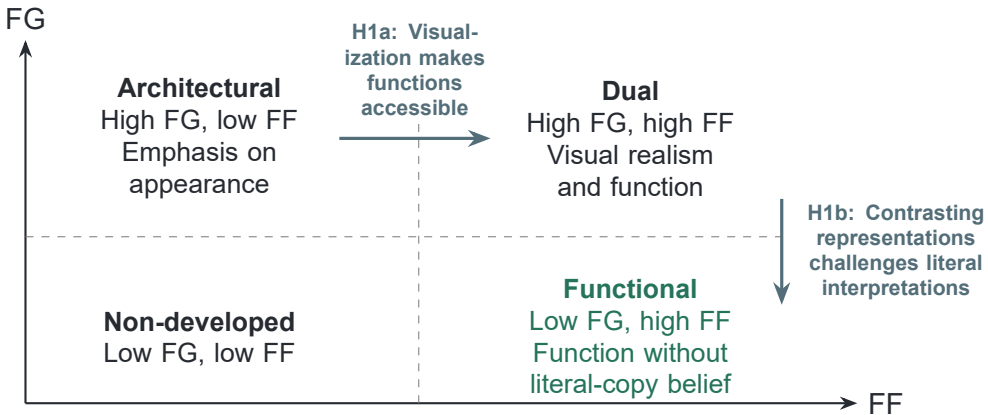
Experimental: Adding Poincaré sphere.



Introductory explanations, guided parameter changes, hints, and solutions.

Hypotheses and study design

Possible changes in model understanding



Why might the sphere change model understanding?

H1a: including the architectural profile

For learners with high FG and low FF, the sphere may make functions accessible: wave plates act as **rotations** of the state, while polarizers perform **projections** of the Jones state.

H1b: including the Dual profile

For learners with high FG and high FF, contrasting the spatial wave representation with the abstract sphere may challenge literal interpretations and reduce FG while FF stays high.

H2: Multimedia effect

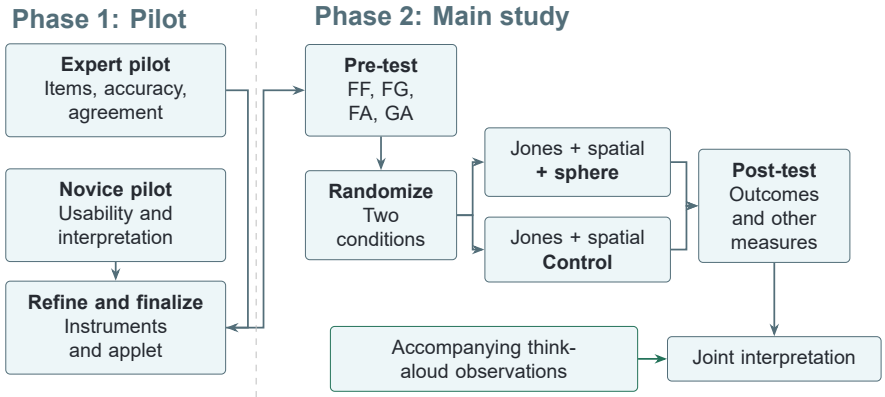
The support with an additional visual representation may increase expert alignment (Rexigel et al., 2024).

The sphere may support both functional reasoning and reflection on visual fidelity.

Hypotheses

	Outcome	Sphere versus control, adjusted for baseline
H1a	Functional Fidelity	Higher FF at post-test.
H1b	Fidelity of Gestalt	Lower FG at post-test.
H2	Expert alignment	Higher FA and GA (structural/Gestalt alignment).

Study structure: pilot, then main study



Phase 1: calibrate the instruments and the applet

Expert pilot

- Classify items as FF, FG, both, or neither.
- Judge factual correctness and the usefulness of analogies.
- Evaluate wording and agreement.

Novice pilot

- Identify confusing controls and instructions.
- Examine interpretation of the representations.
- Assess cognitive load (Krieglstein et al., 2023) and technological acceptance (Abdullah and Ward, 2016).

Refine questionnaire items, applet instructions, and interaction design before the main comparison.

Phase 2: randomized pre/post comparison

Pre-test	FF, FG, FA, and GA
Intervention	Random assignment to guided polarization tasks with or without the added sphere
Post-test	FF, FG, FA, GA; cognitive load and spatial reasoning (Young et al., 2025), demographic information

Population: second- and third-year physics / related STEM undergraduates.

Target: $N = 218$ recruited; minimum analyzable sample $N = 196$.

Think-aloud observations: investigating mechanisms

Students verbalize their thoughts while working through the existing applet tasks, without additional questions or activities.

- Observe how students attempt to incorporate the Poincaré sphere into their mental model.
- Trace difficulties in connecting spatial, symbolic, and sphere representations.
- Examine spontaneously expressed conflicts between literal interpretations and functional relationships.
- Follow how these conflicts are resolved, maintained, or bypassed during task engagement.

Time-ordered observations can provide evidence for mechanisms underlying learning and difficulty.

Bring quantitative outcomes and verbal reasoning together

Quantitative strand

- Estimate condition effects on post-test FF, FG, FA, and GA.
- Adjust for baseline scores and spatial reasoning.
- Explore initial FG, spatial reasoning, and cognitive load.

Qualitative strand

- Trace interpretation, comparison, and revision episodes.
- Examine spontaneously expressed realism-related tension.
- Examine alternative explanations and contrasting cases.

The experiment estimates condition effects. Observations help explain individual learning episodes.

Analysis and current status

Planned primary analysis: MANCOVA

Joint outcomes

$$\mathbf{Y}_{\text{post}} = \begin{pmatrix} FF_{\text{post}} \\ FG_{\text{post}} \\ FA_{\text{post}} \\ GA_{\text{post}} \end{pmatrix}$$

- **Factor:** condition, with or without the additional Poincaré sphere.
- **Covariates:** baseline outcome scores and spatial reasoning ability.
- **Multivariate test:** assess the adjusted condition effect across the four outcomes.
- **Outcome-specific analyses:** examine the directional predictions of H1a, H1b, and H2.

Exploratory analyses: H3 and H4

H3: Spatial reasoning and cognitive load

Regress pre/post changes in the study outcomes on spatial reasoning ability and intrinsic, extraneous, and germane cognitive load.

Do these characteristics help explain variation in learning?

H4: Initial Fidelity of Gestalt as a moderator

Examine the interaction between baseline FG and condition in predicting post-test FF or FA, controlling for the corresponding baseline outcome.

Does the effect of the added visualization depend on students' initial FG?

Initial pilot findings

Pilot studies are ongoing.

Expert pilot

- Add sample and completion status.
- Add item-classification and agreement findings.
- Add resulting item revisions.

Novice pilot

- Add sample and completion status.
- Add interpretation or usability issues.
- Add resulting applet revisions.

Placeholder: pilot figure, applet screenshot, or anonymized spontaneous verbalization

Appendix: The learning sequence

Steps	Topic	Learner activity
1–3	Polarization states	Change amplitudes and relative phase to construct linear and circular states.
4–6	Wave plates	Vary retardance and fast-axis orientation to transform a state.
7	Linear polarizers	Relate transmission-axis orientation to output intensity.
8–9	Combined operators	Coordinate wave plates and polarizers to meet a target.

Nine challenges precede a completion screen and survey return link.

Appendix: measures and protocol decisions

- Cognitive load: Extraneous (5 questions), Intrinsic (5) and Germane (7), 9-point Likert scale (Krieglstein et al., 2023)
- General Extended Technology Acceptance Model for E-Learning (GETAMEL, only in pilot study): (Abdullah and Ward, 2016)
  - Perceived Usefulness
  - Perceived Ease of Use
  - E-Learning Self-Efficacy
  - Perceived Enjoyment
- Spatial reasoning: Measured using the open-source 3D Block Rotation scale from the International Cognitive Ability Resource (ICAR) (Young et al., 2025)

References I

Abdullah, F. and R. Ward (2016). “Developing a General Extended Technology Acceptance Model for E-Learning (GETAMEL) by analysing commonly used external factors”. In: *Computers in Human Behavior* 56, pp. 238–256.

Ainsworth, Shaaron (2006). “DeFT: A conceptual framework for considering learning with multiple representations”. In: *Learning and Instruction* 16.3, pp. 183–198. DOI: 10.1016/j.learninstruc.2006.03.001.

Altshuler, Douglas, Joseph Bahlman, Roslyn Dakin, Andrea Gaede, Benjamin Goller, David Lentink, Paolo Segre, and Dimitri Skandalis (Oct. 2015). “The biophysics of bird flight: functional relationships integrate aerodynamics, morphology, kinematics, muscles, and sensors”. In: *Canadian Journal of Zoology* 93, pp. 961–975. DOI: 10.1139/cjz-2015-0103.

Bley, Jonas (Aug. 28, 2026). *Polarization of Light Applet*. Version 1.0.0.

References II

Krieglstein, Felix, Maik Beege, Günter Daniel Rey, Christina Sanchez-Stockhammer, and Sascha Schneider (2023). "Development and Validation of a Theory-Based Questionnaire to Measure Different Types of Cognitive Load". In: *Educational Psychology Review* 35.1, p. 9. DOI: 10.1007/s10648-023-09738-0.

Mayer, Richard E. (2005). "Cognitive Theory of Multimedia Learning". In: *The Cambridge Handbook of Multimedia Learning*. Ed. by Richard E. Mayer. Cambridge University Press, pp. 31–48. DOI: 10.1017/CBO9780511816819.004.

Rexigel, Eva, Jochen Kuhn, Sebastian Becker, and Sarah Malone (2024). "The More the Better? A Systematic Review and Meta-Analysis of the Benefits of More than Two External Representations in STEM Education". In: *Educational Psychology Review* 36.4, p. 124. DOI: 10.1007/s10648-024-09958-y.

References III

Ubben, Malte S. and Philipp Bitzenbauer (2022). “Two Cognitive Dimensions of Students’ Mental Models in Science: Fidelity of Gestalt and Functional Fidelity”. In: *Education Sciences* 12.3, p. 163. DOI: 10.3390/educsci12030163.

WikimediaCommons (2024). *File:BIRD.jpg — Wikimedia Commons, the free media repository*. [Online; accessed 17-September-2026].

Young, S. R., J. Kim, K. McKee, D. R. Doyle, M. A. Novack, W. Revelle, R. Gershon, and E. M. Dworak (2025). “Validation of International Cognitive Ability Resource (ICAR) Implemented in Mobile Toolbox (MTB)”. In: *Journal of Intelligence* 13.12, p. 154. DOI: 10.3390/jintelligence13120154.